

AMLD Security 3

Plausible Applications of Verified Settlement, Safe Quotienting, and Search Guidance

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Sources

- *Security by Separation, Verification, and Settlement — AMLD Security Treatise v0.1.*
- *What Checks the Proof? — Version 6.80.*

On when the “targets” of automated target provers, automated theorem provers, or automated metalogical deciders are passwords and, more generally, security and vulnerabilities, this document considers how AMLD-style reasoning could plausibly be used defensively. The central idea is not that a search system should be trusted because it is powerful, but that powerful search should be separated from the mechanism that certifies what may enter trusted state.

The discussion below preserves the application categories from the working document. Under each *Plausible application* heading, the informal prose has been replaced by explicit LaTeX-ready mathematical formulations.

1 Cryptography

Discovering unexpected mathematical structure in a supposedly hard problem could undermine an encryption scheme. The dramatic case would be a new efficient attack on assumptions behind widely deployed public-key cryptography. This would not imply that “all encryption is broken”; distinct constructions rest on distinct assumptions.

Plausible application

Let a cryptographic search process generate histories $h \in \mathcal{H}$, with continuation operator $\mathcal{T}(h, a)$ for action $a \in A(h)$. A quotient candidate

$$Q : \mathcal{H} \rightarrow \mathcal{S}$$

is *future-sufficient* only if

$$Q(h_1) = Q(h_2) \implies \left(\text{Future}(h_1) \equiv_{\text{Ver}} \text{Future}(h_2) \right),$$

where $\equiv_{\text{Ver}}$ means equivalence with respect to every fact required by later search, scope, provenance, and certificate verification. Thus a safe speedup is not “try keys faster” but a proved reduction

$$\mathcal{H} / \sim_Q, \quad h_1 \sim_Q h_2 \iff Q(h_1) = Q(h_2),$$

such that certified solutions lift:

$$\text{Ver}(\pi, [h]_Q) = 1 \implies \exists h' \in [h]_Q \text{ Ver}(\pi, h') = 1.$$

A COMPASS-type policy may rank candidate transformations by

$$S(a \mid h) = w_1 \text{sim}(a, h) + w_2 \rho_Q(a, h) + w_3 d(a, h) - w_4 \widehat{\text{Cost}}(a \mid h),$$

but S is a policy score, not a truth predicate. The soundness boundary remains

$$\text{promotion to BANK} \iff \text{Ver}(\pi, \text{claim}, \Gamma) = 1.$$

Hence a cryptographically significant shortcut would require a certificate that the quotient or invariant preserves the target problem. No such formalism alone establishes an attack on AES, RSA, ECC, or post-quantum schemes.

2 Cybersecurity and Vulnerability Research

A strong reasoning system could search for overlooked weaknesses, but the defensive use is to turn vulnerability discovery into a certificate-producing audit problem and to separate speculative traces from verified defects.

Plausible application

Let Z be the set of complete execution traces. Define

$$\text{Impl}(z) \in \{0, 1\}, \quad \text{Req}(z) \in \{0, 1\},$$

and the acceptance-soundness property

$$\Phi_{\text{safe}} := \forall z \in Z [\text{Impl}(z) \Rightarrow \text{Req}(z)].$$

A single checked witness z^* satisfying

$$\text{Impl}(z^*) = 1, \quad \text{Req}(z^*) = 0$$

is a finite refutation certificate:

$$z^* \in \mathcal{R}(\Phi_{\text{safe}}).$$

An AMLD settlement may therefore be modeled as

$$\Sigma(\Phi) \in \{\mathcal{P}, \mathcal{R}, \mathcal{I}, \mathcal{C}, \text{UNKNOWN}\},$$

where P searches for a proof of Φ , R for a checked counterexample, I for independence relative to a declared metatheory, and C for inconsistency of the admitted specification. Candidate traces u may enter FUTUREBANK_t , but trusted state evolves only by

$$\text{BANK}_{t+1} = \text{BANK}_t \cup \{u \in \text{FUTUREBANK}_t : \text{Ver}(u, \Gamma_t) = 1\}.$$

For candidate analysis machinery M , define an Ocean gate

$$M \rightsquigarrow \text{BANK} \quad \text{only if} \quad \forall o \in \mathcal{O}, \text{Acc}(M, o) = 1.$$

Failures discovered during this gate become persistent regression obligations rather than discarded search debris.

3 Critical Infrastructure

Hospitals, power systems, water utilities, transportation networks, and communications systems require containment because a cyber failure can become a physical failure. Here the natural object is a compromise graph rather than a single “secure/insecure” bit.

Plausible application

Model an infrastructure as a directed graph

$$G = (V, E), \quad \text{Reach}_G(c) = \{v \in V : c \rightsquigarrow_G v\}.$$

The blast radius of compromise at c is

$$B_G(c) := \text{Reach}_G(c).$$

If segmentation produces $G' = (V, E')$ with $E' \subseteq E$, then monotonicity of reachability gives

$$B_{G'}(c) \subseteq B_G(c), \quad |B_{G'}(c)| \leq |B_G(c)|.$$

A defensive controller should therefore optimize a vector, not one scalar:

$$\mathbf{J}(D) = (p_{\text{comp}}(D), \max_c |B_{G_D}(c)|, A_{\text{loss}}(D), C_{\text{verify}}(D), C_{\text{repair}}(D)).$$

A design D_2 dominates D_1 for vulnerability class X only relative to declared non-regression constraints:

$$D_2 \succ_X D_1 \iff m_X(D_2) < m_X(D_1) \wedge \bigwedge_{j \in J_{\text{protected}}} m_j(D_2) \leq m_j(D_1).$$

This lets an AMLD compare segmentation strategies by certificate-checked reachability bounds while penalizing defenses whose availability cost is unacceptable. In a hospital or utility, a defense that reduces attack reach but prevents legitimate operation is not automatically Pareto-superior.

4 Military and Intelligence Systems

Secure communications, cryptanalysis, compartmentation, logistics, and autonomous-system verification all require strong separation between speculative reasoning and trusted authority.

Plausible application

Let the system be partitioned into security domains

$$V = \bigsqcup_{i=1}^k V_i$$

with boundary-crossing transitions

$$e = (x, y) \in E, \quad x \in V_i, y \in V_j, i \neq j.$$

Associate to each crossing a policy predicate $P_e(s, \alpha)$, where s is system state and α the asserted authority. An admissible transition system satisfies

$$\forall e \in E_{\text{cross}} \forall (s, \alpha), \quad \text{Fire}(e, s, \alpha) \Rightarrow P_e(s, \alpha).$$

The corresponding security target is a universal invariant

$$\Phi_{\text{sep}} = \Box(\text{Cross}(s, s') \Rightarrow \text{Authorized}(s, s')).$$

AMLD may settle Φ_{sep} by a proof certificate, a checked violating trace, a declared independence certificate, or an inconsistency certificate:

$$\Sigma(\Phi_{\text{sep}}) \in \{P, R, I, C, \text{UNKNOWN}\}.$$

Crucially, UNKNOWN is not evidence of safety. Search authority and operational authority are separated by requiring

$$\text{Ver}(\pi, \Phi, \Gamma) = 1 \quad \text{before} \quad \Phi \in \text{BANK}.$$

This formalizes compartmentation as a reachability-and-authorization problem without granting a creative reasoning component direct authority to change trusted operational state.

5 Software Correctness

Safety-critical software can be treated as a transition system whose invariants are proved or refuted by machine-checkable certificates.

Plausible application

Let

$$\mathcal{M} = (S, S_0, \rightarrow)$$

be a transition system and $I : S \rightarrow \{0, 1\}$ a safety invariant. Inductiveness requires

$$\forall s \in S_0, I(s) = 1, \quad \forall s, s' \in S, (I(s) \wedge s \rightarrow s') \Rightarrow I(s').$$

Then

$$\forall s \in \text{Reach}_{\mathcal{M}}(S_0), I(s) = 1.$$

Role P searches for a proof of the base and step obligations, while R searches for a finite trace

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_n$$

with $s_0 \in S_0$ and $I(s_n) = 0$. The AMLD architecture itself can be subject to executable invariants, for example

$$u \in \text{BANK}_{t+1} \setminus \text{BANK}_t \implies \exists r \text{ Ver}(r, u) = 1,$$

$$\text{Ver}(r, u) = 0 \implies u \notin \text{BANK}_{t+1},$$

and, for dependency set $\text{Dep}(u)$,

$$u \in \text{BANK}_t \implies \text{Dep}(u) \subseteq \text{BANK}_t.$$

These conditions make verifier-gated promotion, dependency closure, and rejection persistence software properties that can be regression-tested. Learned guidance may choose proof obligations, but only the checker determines whether the invariant has been established.

6 Scientific Discovery

The Quotient Hunter idea is especially relevant when a small invariant can replace an enormous search space by an equivalent quotient.

Plausible application

Let a scientific model generate histories $h \in \mathcal{H}$ and observables

$$Y_h(\tau), \quad \tau \geq 0.$$

A descriptor $Q : \mathcal{H} \rightarrow \mathcal{S}$ is future-sufficient for observable family $\mathcal{F}$ when

$$Q(h_1) = Q(h_2) \implies \forall f \in \mathcal{F} \forall \tau \geq 0, \quad f(Y_{h_1}(\tau)) = f(Y_{h_2}(\tau)),$$

or, in stochastic settings, equality is replaced by equality of the relevant conditional law:

$$\mathcal{L}(Y_{\text{future}} \mid h) = \mathcal{L}(Y_{\text{future}} \mid Q(h)).$$

Then inference may be performed on the quotient state space

$$\mathcal{S} \simeq \mathcal{H} / \sim_Q$$

rather than on every history separately. FUTUREBANK can retain conjectural Q 's and candidate invariants, while a ranking policy allocates compute by an estimated information/compression objective such as

$$U(Q) = \hat{I}(Q; Y_{\text{future}}) - \lambda \log |\text{im } Q| - \mu \widehat{\text{Cost}}(Q).$$

However, $U(Q)$ only prioritizes candidates. Promotion requires a proof, verified empirical certificate, or other domain-appropriate checker establishing the preservation claim. The scientific payoff is therefore a certified reduction of effective state dimension, not merely a correlation found by a heuristic.

7 Engineering Optimization

Circuit design, scheduling, placement, routing, materials design, and supply-chain optimization all contain large combinatorial spaces where safe state merging could reduce repeated downstream search.

Plausible application

Let an engineering problem be

$$\min_{x \in X} f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_j(x) = 0, \quad j = 1, \dots, r.$$

For a partial design history h , define its continuation set

$$\mathcal{C}(h) = \{x \in X : x \text{ extends } h \text{ and satisfies all inherited constraints}\}.$$

A safe quotient Q must preserve downstream feasibility and objective information:

$$Q(h_1) = Q(h_2) \implies \left(\inf_{x \in \mathcal{C}(h_1)} f(x) = \inf_{x \in \mathcal{C}(h_2)} f(x) \right)$$

together with an explicit correspondence between feasible completions. A COMPASS-like scheduler may rank expansions by

$$S(a \mid h) = \alpha \Delta_{\text{goal}} + \beta \rho_Q + \gamma \text{reuse} - \delta \widehat{\text{Cost}} - \eta \widehat{\text{Risk}},$$

while feasibility remains checker-defined:

$$\text{Accept}(x) \iff \bigwedge_i g_i(x) \leq 0 \wedge \bigwedge_j h_j(x) = 0 \wedge \text{Ver}(x) = 1.$$

Thus learned search can decide *where* to spend computation, while verified constraints decide *what counts* as an admissible design. Safe quotienting is useful only when future constraints are provably preserved.

8 AI Safety Itself

A shortcut-generating system is materially safer when creative search is separated from the authority to declare a result valid or promote it into trusted state.

Plausible application

Let G_θ be a mutable generator/search policy and V a fixed or separately governed verifier. For candidate claim c with certificate π , define

$$\text{Promote}(c, \pi) \iff V(c, \pi, \Gamma) = 1.$$

Policy-neutral verification requires

$$\forall \theta_1, \theta_2, c, \pi, \Gamma, \quad V_{\theta_1}(c, \pi, \Gamma) = V_{\theta_2}(c, \pi, \Gamma) = V(c, \pi, \Gamma),$$

so changing the search policy does not silently change truth conditions. Speculative self-modifications θ' remain outside trusted state:

$$\theta' \in \text{FUTUREBANK}_t \not\Rightarrow \theta' \in \text{BANK}_t.$$

Instead, candidate versions are evaluated against an Ocean regression set

$$\mathcal{O}_t = \{o_1, \dots, o_n\}, \quad R(\theta') = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{Acc}(G_{\theta'}, o_i) = 1],$$

plus hard invariants $H_j(\theta')$. Promotion can require

$$R(\theta') \geq \tau \quad \wedge \quad \bigwedge_j H_j(\theta') = 1 \quad \wedge \quad V(\pi_{\theta'}, \theta', \Gamma) = 1.$$

This architecture does not prove that a powerful reasoning system is globally safe. It does provide a precise technical separation between generation, testing, verification, and authority, while retaining failed proposals as future regression evidence.

Research Boundary

These constructions are research hypotheses and formal design patterns. They describe how AMLD-style settlement, safe quotienting, BANK/FUTUREBANK separation, Ocean regression gates, and policy-neutral verification could be used to structure defensive reasoning. They do not constitute evidence that any particular cryptographic primitive, deployed infrastructure, or software system is vulnerable.

Source anchors

The mathematical motifs above are based on the working sources cited in the original document: *What Checks the Proof?* v6.80, especially its discussions of policy-neutral verification, FUTUREBANK, COMPASS, software invariants, and safe quotienting; and *Security by Separation, Verification, and Settlement*, especially its discussions of separation, blast radius, security conformance, P/R/I/C settlement, the Ocean gate, and multi-objective security.